

Summary of bug information

Benchmark	Bug Type	Number of Bug
Netcdf	Assertion_Failure	1
	Double_Free	1
	Heap_Corruption	2
Total		4

[1] Subject	Netcdf-ncdump
Vulnerability Type	Assertion_Failure
Input	PoC/Netcdf/ncdump/Assertion_Failure_01
Description	Run the program with the input, program crashed due to Assert_Fail Command Line: ./ncdump -b f Assertion_failure_01
Stack Info	<pre> (gdb) ? Starting program: /root/.graphdissect/benchmarks/netcdf/ncdump/ncdump -b f /root/.graphdissect/benchmarks/netcdf/ncdump/SIGABRT.PC.7d6d3406d9fc.STACK.ca471260e.CODE.-6.ADOR.0.INSTR.m 0x.....hex\sr13d fi2z warning: Error disabling address space randomization: Operation not permitted [Thread debugging using libthread_db enabled] Using host libthread_db library "/lib64/libthread_db.so.1". netcdf SIGABRT.PC.7d6d3406d9fc.STACK.ca471260e.CODE.-6.ADOR.0.INSTR.mov.....hex\sr13d { dimensions: ncdump malloc.c:2617: sysmalloc: Assertion 'old_top == initial_top (av) && old_size == 0' { ((unsigned long) (old_size) >= MINSIZE && prev_inuse (old_top) && ((unsigned long) ol d_end & (pagesize - 1)) == 0) failed. Program received signal SIGABRT, Aborted. __pthread_kill_implementation (no_tid=0, signo=6, threadid=124814488299456) at ./nptl/pthread_kill.c:44 44 ./nptl/pthread_kill.c: No such file or directory. (gdb) bt #0 __pthread_kill_implementation (no_tid=0, signo=6, threadid=124814488299456) at ./nptl/pthread_kill.c:44 #1 __pthread_kill_internal (signo=6, threadid=124814488299456) at ./nptl/pthread_kill.c:78 #2 __GI___pthread_kill (threadid=124814488299456, signo=signoentry=6) at ./nptl/pthread_kill.c:89 #3 0x00007184a54547f3 in __GI_raise (sig=signoentry=6) at ./sysdeps/posix/raise.c:26 #4 0x00007184a54547f3 in __GI_abort () at ./stdlib/abort.c:79 #5 0x00007184a54547e5 in __malloc_assert () assertion=assertionentry=0x7184a5621988 "old_top == initial_top (av) && old_size == 0' { ((unsigned long) (old_size) >= MINSIZE && prev_inuse (old_top) && ((unsigned long) o ld_end & (pagesize - 1)) == 0) ", file=fileentry=0x7184a561c68c "malloc.c", line=lineentry=0x7184a56221f8 <__PRETTY_FUNCTION__ > "sysmalloc") at ./malloc/malloc.c:387 #6 0x00007184a5454937 in sysmalloc (nb=nbentry=32, av=aventry=0x7184a565dc80 <main_arena>) at ./malloc/malloc.c:2617 #7 0x00007184a54547d6 in __ut_malloc (b=aventry=0x7184a565dc80 <main_arena>, bytes=bytesentry=1) at ./malloc/malloc.c:4407 #8 0x00007184a5454262 in __GI___libc_malloc (bytes=1) at ./malloc/malloc.c:3321 #9 0x0000000000000221 in escaped_name () #10 0x00000000000004f108 in print_name () #11 0x0000000000000439076 in dn_ncdump_rec () #12 0x0000000000000372e7 in dn_ncdump () #13 0x00000000000001360ad in main () (gdb) </pre>
PoC	PoC/Netcdf/ncdump
Status	PoC Done, issue submitted Heap allocation crash in escaped_name() when running ncdump -b f on a NetCDF file · Issue #3311 · Unidata/netcdf-c

[4] Subject	Netcdf-ncdump
Vulnerability Type	Heap_Corruption
Input	PoC/Netcdf/ncdump/Heap_Corruption_02
Description	Run the program with the input, program crashed due to heap corruption Command Line: ./ncdump -b f Heap_Corruption_02
Stack Info	<pre> (gdb) r Starting program: /root/.gdbinit/benchmarks/netcdf/ncdump/ncdump -b f /root/.gdbinit/benchmarks/netcdf/ncdump/SIGABRT.PC.753363dd79fc.STACK.1834415f35.CODE.-6.ADR.0.INSTR. mov __hex,0x13d,fuzz warning: Error disabling address space randomization: Operation not permitted [Thread debugging using libthread_db enabled] Using host libthread_db library "/usr/lib64/libthread_db.so.1". netcdf SIGABRT.PC.753363dd79fc.STACK.1834415f35.CODE.-6.ADR.0.INSTR.mov __hex,\r13d { dimensions: = UNLIMITED ; // (0 currently) free(): invalid pointer Program received signal SIGABRT, Aborted. __pthread_kill_implementation (no_tid=0, signo=6, threadid=131432709384128) at ./nptl/pthread_kill.c:144 144 ./nptl/pthread_kill.c: No such file or directory. (gdb) bt #0 __pthread_kill_implementation (no_tid=0, signo=6, threadid=131432709384128) at ./nptl/pthread_kill.c:144 #1 __pthread_kill_internal (signo=6, threadid=131432709384128) at ./nptl/pthread_kill.c:78 #2 __GI___pthread_kill (threadid=131432709384128, signo=signoentry4) at ./nptl/pthread_kill.c:89 #3 0x00007f8951ff0476 in __GI_raise (sig=signoentry6) at ./sysdeps/posix/raise.c:26 #4 0x00007f8951fa37f3 in __GI_abort () at ./stdlib/abort.c:79 #5 0x00007f895204a077 in __libc_message (action=actionentry40, abort, fmt=fmtentry=0x7f895219d777 "bain") at ./sysdeps/posix/libc_fatal.c:156 #6 0x00007f895205bcfc in malloc_printerr (str=stringentry=0x7f895219d744 "free(): invalid pointer") at ./malloc/malloc.c:5664 #7 0x00007f8952050414 in _int_free (p=optimized out, p=optimized out, have_lock=0) at ./malloc/malloc.c:4439 #8 0x00007f8952050453 in __GI___libc_free (mem=optimized out) at ./malloc/malloc.c:3391 #9 0x0000000000439c07 in do_ncdump_rec () #10 0x00000000004370e7 in do_ncdump () #11 0x0000000000436ba8 in main () (gdb) </pre>
PoC	PoC/Netcdf/ncdump
Status	PoC Done, issue submitted Invalid free in do_ncdump_rec() when running ncdump -b f on a NetCDF file · Issue #3313 · Unidata/netcdf-c